预取流程

D:\ProgramData\anaconda3\envs\concrete-env\python.exe
E:\GithubProjects\ckks_fed_robustness\main.py

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════ CKKS-Fed: 基于CKKS同态加密的联邦学习框架 ═══════════════════════════════════

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils: 🚀 启动 CipherFetch 密文预取实验（FedFetch 风格）

INFO:ckks_fed.log_utils:📦 Prefetch Experiment Initialization

INFO:ckks_fed.log_utils:[初始化/INIT] Dataset loaded: mnist | train=60000 | test=10000

INFO:ckks_fed.log_utils:[初始化/INIT] Non-IID partition (α=0.5): samples/client=[5632, 6704, 4496, 2848, 6997, 9951, 6406, 6719, 4097, 6150]

INFO:ckks_fed.data:

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Client Data Distribution

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Client 0: total= 5632 | 0: 8 1: 4 2: 802 3: 221 4:2606 5: 55 6:1406 7: 64 8: 330 9: 136

Client 1: total= 6704 | 0: 93 1: 57 2: 129 3:1764 4:1557 5: 6 6: 105 7: 898 8:2095 9: 0

Client 2: total= 4496 | 0: 230 1:1027 2:1774 3: 586 4: 33 5: 430 6: 16 7: 154 8: 216 9: 30

Client 3: total= 2848 | 0: 442 1: 197 2: 1 3: 0 4: 571 5: 430 6: 531 7: 16 8: 544 9: 116

Client 4: total= 6997 | 0: 174 1:2219 2:1029 3: 278 4: 753 5: 478 6: 10 7: 397 8: 31 9:1628

Client 5: total= 9951 | 0:1213 1: 633 2: 3 3: 64 4: 201 5:3589 6: 434 7:3064 8: 578 9: 172

Client 6: total= 6406 | 0: 422 1:2177 2: 11 3:1633 4: 31 5: 32 6: 853 7: 1 8: 938 9: 308

Client 7: total= 6719 | 0:1086 1: 48 2: 383 3:1452 4: 4 5: 165 6: 165 7: 170 8: 47 9:3199

Client 8: total= 4097 | 0:1251 1: 347 2: 3 3: 95 4: 17 5: 3 6: 656 7:1366 8: 47 9: 312

Client 9: total= 6150 | 0:1004 1: 33 2:1823 3: 38 4: 69 5: 233 6:1742 7: 135 8:1025 9: 48

Per-class total: 0: 5923 1: 6742 2: 5958 3: 6131 4: 5842 5: 5421 6: 5918 7: 6265 8: 5851 9: 5949

Overall total samples: 60000

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INFO:ckks_fed.log_utils: └── 🗂️ Dataset: mnist | Clients: 10 | IID=False

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Model: LeNet

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Layer (type)	Output Shape	Param #
Conv2d-1	[-1,6,24,24]	156
ReLU-2	[-1,6,24,24]	0
MaxPool2d-3	[-1,6,12,12]	0
Conv2d-4	[-1,16,8,8]	2,416
ReLU-5	[-1,16,8,8]	0
MaxPool2d-6	[-1,16,4,4]	0
Flatten-7	[-1,256]	0
Linear-8	[-1,120]	30,840
ReLU-9	[-1,120]	0
Linear-10	[-1,84]	10,164
ReLU-11	[-1,84]	0
Linear-12	[-1,10]	850

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Total params: 44,426

Trainable params: 44,426

Non-trainable params: 0

Model size (float32): 0.17 MB

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INFO:ckks_fed.crypto:CKKS上下文创建成功: N=8192, mod_bits=[40, 20, 40], scale=1048576

INFO:ckks_fed.crypto:CKKS管理器初始化完成 (slot_count=4096, slot_packing=ON)

INFO:ckks_fed.log_utils: └── 🔐 CKKS unified context | N=8192 | scale=2^20 | serialized=375.33 KB

INFO:ckks_fed.log_utils: └── 🌐 Bandwidth Simulator | DL=0.0 kbps | UL=0.0 kbps | dist=uniform

INFO:ckks_fed.log_utils:[调度/SCHED] Delta size = 2.85 MB per round

INFO:ckks_fed.log_utils: └── 📅 Prefetch Scheduler | window=3 | warmup=2 | β=0.5 | OC=1.0

INFO:ckks_fed.log_utils:[服务器/SERVER] Initialized | rule=fedavg | filter=None

INFO:ckks_fed.log_utils:[服务器/SERVER] PrefetchServer init | enable=True | window=3 | warmup=2

INFO:ckks_fed.log_utils: └── 🖥️ Prefetch Aggregation Server

INFO:ckks_fed.log_utils:[客户端/CLIENT] #0 initialized | data_size=5632

INFO:ckks_fed.log_utils:[客户端/CLIENT] #0 PrefetchClient init | enable=True

INFO:ckks_fed.crypto:CKKS管理器初始化完成 (slot_count=4096, slot_packing=ON)

INFO:ckks_fed.log_utils:[客户端/CLIENT] #1 initialized | data_size=6704

INFO:ckks_fed.log_utils:[客户端/CLIENT] #1 PrefetchClient init | enable=True

INFO:ckks_fed.crypto:CKKS管理器初始化完成 (slot_count=4096, slot_packing=ON)

INFO:ckks_fed.log_utils:[客户端/CLIENT] #2 initialized | data_size=4496

INFO:ckks_fed.log_utils:[客户端/CLIENT] #2 PrefetchClient init | enable=True

INFO:ckks_fed.crypto:CKKS管理器初始化完成 (slot_count=4096, slot_packing=ON)

INFO:ckks_fed.log_utils:[客户端/CLIENT] #3 initialized | data_size=2848

INFO:ckks_fed.log_utils:[客户端/CLIENT] #3 PrefetchClient init | enable=True

INFO:ckks_fed.crypto:CKKS管理器初始化完成 (slot_count=4096, slot_packing=ON)

INFO:ckks_fed.log_utils:[客户端/CLIENT] #4 initialized | data_size=6997

INFO:ckks_fed.log_utils:[客户端/CLIENT] #4 PrefetchClient init | enable=True

INFO:ckks_fed.crypto:CKKS管理器初始化完成 (slot_count=4096, slot_packing=ON)

INFO:ckks_fed.log_utils:[客户端/CLIENT] #5 initialized | data_size=9951

INFO:ckks_fed.log_utils:[客户端/CLIENT] #5 PrefetchClient init | enable=True

INFO:ckks_fed.crypto:CKKS管理器初始化完成 (slot_count=4096, slot_packing=ON)

INFO:ckks_fed.log_utils:[客户端/CLIENT] #6 initialized | data_size=6406

INFO:ckks_fed.log_utils:[客户端/CLIENT] #6 PrefetchClient init | enable=True

INFO:ckks_fed.crypto:CKKS管理器初始化完成 (slot_count=4096, slot_packing=ON)

INFO:ckks_fed.log_utils:[客户端/CLIENT] #7 initialized | data_size=6719

INFO:ckks_fed.log_utils:[客户端/CLIENT] #7 PrefetchClient init | enable=True

INFO:ckks_fed.crypto:CKKS管理器初始化完成 (slot_count=4096, slot_packing=ON)

INFO:ckks_fed.log_utils:[客户端/CLIENT] #8 initialized | data_size=4097

INFO:ckks_fed.log_utils:[客户端/CLIENT] #8 PrefetchClient init | enable=True

INFO:ckks_fed.crypto:CKKS管理器初始化完成 (slot_count=4096, slot_packing=ON)

INFO:ckks_fed.log_utils:[客户端/CLIENT] #9 initialized | data_size=6150

INFO:ckks_fed.log_utils:[客户端/CLIENT] #9 PrefetchClient init | enable=True

INFO:ckks_fed.crypto:CKKS管理器初始化完成 (slot_count=4096, slot_packing=ON)

INFO:ckks_fed.log_utils: └── 👤 Created 10 prefetch clients

INFO:ckks_fed.log_utils:[粘采样/STICKY] Initialized | mode=UNIFORM | clients=10 | OC=1.0 | decay=0.9

INFO:ckks_fed.log_utils: └── 🎲 Presampling (UNIFORM) | OC=1.0 | decay=0.9

INFO:ckks_fed.log_utils:[初始化/INIT] Experiment output dir: ./results\experiment_20260420_182609

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════════ CipherFetch 密文预取实验启动 ═══════════════════════════════════════

INFO:ckks_fed.log_utils: 总轮数: 20

INFO:ckks_fed.log_utils: 上传模式: 完整权重(Full)

INFO:ckks_fed.log_utils:═══

INFO:ckks_fed.log_utils:[初始化/INIT] 统一 CKKS 密钥已分发给 10 个客户端（含私钥，用于本地解密）

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════ 第 1/20 轮 | CipherFetch 预取 ═══════════════════════════════════

INFO:ckks_fed.log_utils:[粘采样/STICKY] Select | target=0 | no pool, fallback random=[8, 1, 5, 0, 7]

INFO:ckks_fed.log_utils:[选择/SELECT] Presampled-Current 选中客户端: [8, 1, 5, 0, 7] (共5/10个)

INFO:ckks_fed.log_utils:[预取/PREFETCH] Warmup 阶段：跳过预取广播（round 1 < warmup 3）

INFO:ckks_fed.log_utils: 💻 本地训练阶段（Fetch → Apply → Compute → Upload）

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#8 | 损失=1.2207 | 准确率=13.02% | 耗时=2.48s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#1 | 损失=1.1568 | 准确率=12.49% | 耗时=3.09s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#5 | 损失=0.8949 | 准确率=14.63% | 耗时=5.39s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#0 | 损失=1.2181 | 准确率=11.71% | 耗时=3.23s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#7 | 损失=1.1077 | 准确率=13.04% | 耗时=3.81s | 上传=完整权重

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#8 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#1 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#5 | 加密耗时=0.04s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#0 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#7 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　Expansion 16.81x | plain=867.70 KB | cipher=14.25 MB

INFO:ckks_fed.log_utils:[聚合/AGG]　　Aggregating 5 encrypted updates...

INFO:ckks_fed.log_utils:[聚合/AGG]　　参与=5人 | 保留=5人 | 规则=CKKS+fedavg | 聚合耗时=0.033s | 上传总量=14.25 MB | 下载总量=867.70 KB

INFO:ckks_fed.log_utils:[服务器/SERVER]　全局模型已更新 | 当前轮次=1 | 上传模式=完整权重替换

INFO:ckks_fed.log_utils:[服务器/SERVER]　密文增量已准备 | 大小=2.85 MB | 加密耗时=0.0706s

INFO:ckks_fed.log_utils:[评估/EVAL]　　测试损失=1.6622 | 测试准确率=46.38%

INFO:ckks_fed.log_utils:[轮次/ROUND]　#1 done | acc=46.38% | round_dur=66.96s | vclock=1.12 min | comm=15.09 MB

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════ 第 2/20 轮 | CipherFetch 预取 ═══════════════════════════════════

INFO:ckks_fed.log_utils:[粘采样/STICKY] Select | target=1 | no pool, fallback random=[0, 1, 8, 5, 3]

INFO:ckks_fed.log_utils:[选择/SELECT] Presampled-Current 选中客户端: [0, 1, 8, 5, 3] (共5/10个)

INFO:ckks_fed.log_utils:[预取/PREFETCH] Warmup 阶段：跳过预取广播（round 2 < warmup 3）

INFO:ckks_fed.log_utils: 💻 本地训练阶段（Fetch → Apply → Compute → Upload）

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#0 | 损失=0.3638 | 准确率=17.78% | 耗时=3.17s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#1 | 损失=0.3214 | 准确率=18.04% | 耗时=3.86s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#8 | 损失=0.3922 | 准确率=17.94% | 耗时=2.40s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#5 | 损失=0.2539 | 准确率=18.50% | 耗时=5.70s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#3 | 损失=0.5120 | 准确率=16.89% | 耗时=1.70s | 上传=完整权重

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#0 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#1 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#8 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#5 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#3 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] Expansion 16.82x | plain=867.70 KB | cipher=14.25 MB

INFO:ckks_fed.log_utils:[聚合/AGG] Aggregating 5 encrypted updates...

INFO:ckks_fed.log_utils:[聚合/AGG] 参与=5人 | 保留=5人 | 规则=CKKS+fedavg | 聚合耗时=0.033s | 上传总量=14.25 MB | 下载总量=867.70 KB

INFO:ckks_fed.log_utils:[服务器/SERVER] 全局模型已更新 | 当前轮次=2 | 上传模式=完整权重替换

INFO:ckks_fed.log_utils:[服务器/SERVER] 密文增量已准备 | 大小=2.85 MB | 加密耗时=0.0757s

INFO:ckks_fed.log_utils:[评估/EVAL] 测试损失=0.8151 | 测试准确率=72.46%

INFO:ckks_fed.log_utils:[轮次/ROUND] #2 done | acc=72.46% | round_dur=67.00s | vclock=2.23 min | comm=15.10 MB

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════ 第 3/20 轮 | CipherFetch 预取 ═══════════════════════════════════

INFO:ckks_fed.log_utils:[粘采样/STICKY] Select | target=2 | no pool, fallback random=[9, 2, 0, 6, 8]

INFO:ckks_fed.log_utils:[选择/SELECT] Presampled-Current 选中客户端: [9, 2, 0, 6, 8] (共5/10个)

INFO:ckks_fed.log_utils: 🔮 预取调度：为第 6 轮提前安排下载计划

INFO:ckks_fed.log_utils:[粘采样/STICKY] Presample | target=5 | sampled=5 (need=5, OC=1.0) | ids=[1, 7, 6, 2, 8]

INFO:ckks_fed.log_utils:[调度/SCHED] 当前轮=2 目标轮=5 | 最早启动轮=r_min(2) 可延后=0人 需立即启动=5人 | (OC=1.0, β=0.5)

INFO:ckks_fed.log_utils:[预取/PREFETCH] Broadcast → 5 clients | targets=[1, 3, 4, 5, 7]

INFO:ckks_fed.log_utils:[预取/PREFETCH] 广播完成 | 2 个空闲客户端已累加密文增量 | 平均缓存 1 个 delta (池内 2/5)

INFO:ckks_fed.log_utils: 💻 本地训练阶段（Fetch → Apply → Compute → Upload）

INFO:ckks_fed.log_utils:[客户端/CLIENT] #9 apply_prefetch | deltas=1 | apply_time=0.0177s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#9 | 损失=0.1433 | 准确率=19.13% | 耗时=3.89s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #2 apply_prefetch | deltas=1 | apply_time=0.0162s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#2 | 损失=0.1883 | 准确率=18.84% | 耗时=2.78s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#0 | 损失=0.1945 | 准确率=18.74% | 耗时=3.25s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #6 apply_prefetch | deltas=1 | apply_time=0.0179s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#6 | 损失=0.1394 | 准确率=19.19% | 耗时=3.80s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#8 | 损失=0.1506 | 准确率=19.04% | 耗时=2.39s | 上传=完整权重

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#9 | 加密耗时=0.04s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#2 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#0 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#6 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#8 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] Expansion 16.81x | plain=867.70 KB | cipher=14.25 MB

INFO:ckks_fed.log_utils:[聚合/AGG] Aggregating 5 encrypted updates...

INFO:ckks_fed.log_utils:[聚合/AGG] 参与=5人 | 保留=5人 | 规则=CKKS+fedavg | 聚合耗时=0.034s | 上传总量=14.25 MB | 下载总量=867.70 KB

INFO:ckks_fed.log_utils:[服务器/SERVER] 全局模型已更新 | 当前轮次=3 | 上传模式=完整权重替换

INFO:ckks_fed.log_utils:[服务器/SERVER] 密文增量已准备 | 大小=2.85 MB | 加密耗时=0.0763s

INFO:ckks_fed.log_utils:[评估/EVAL] 测试损失=0.2738 | 测试准确率=91.43%

INFO:ckks_fed.log_utils:[轮次/ROUND] #3 done | acc=91.43% | round_dur=68.68s | vclock=3.38 min | comm=15.09 MB

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════ 第 4/20 轮 | CipherFetch 预取 ═══════════════════════════════════

INFO:ckks_fed.log_utils:[粘采样/STICKY] Select | target=3 | no pool, fallback random=[1, 5, 4, 8, 0]

INFO:ckks_fed.log_utils:[选择/SELECT] Presampled-Current 选中客户端: [1, 5, 4, 8, 0] (共5/10个)

INFO:ckks_fed.log_utils: 🔮 预取调度：为第 7 轮提前安排下载计划

INFO:ckks_fed.log_utils:[粘采样/STICKY] Presample | target=6 | sampled=5 (need=5, OC=1.0) | ids=[8, 2, 5, 7, 3]

INFO:ckks_fed.log_utils:[调度/SCHED] 当前轮=3 目标轮=6 | 最早启动轮=r_min(3) 可延后=0人 需立即启动=5人 | (OC=1.0, β=0.5)

INFO:ckks_fed.log_utils:[预取/PREFETCH] Broadcast → 5 clients | targets=[2, 3, 6, 7, 9]

INFO:ckks_fed.log_utils:[预取/PREFETCH] 广播完成 | 3 个空闲客户端已累加密文增量 | 平均缓存 1 个 delta (池内 3/5)

INFO:ckks_fed.log_utils: 💻 本地训练阶段（Fetch → Apply → Compute → Upload）

INFO:ckks_fed.log_utils:[客户端/CLIENT] #1 apply_prefetch | deltas=1 | apply_time=0.0187s

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#1 | 损失=0.1094 | 准确率=19.38% | 耗时=4.09s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #5 apply_prefetch | deltas=1 | apply_time=0.0158s

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#5 | 损失=0.0779 | 准确率=19.53% | 耗时=5.87s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #4 apply_prefetch | deltas=2 | apply_time=0.0157s

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#4 | 损失=0.1371 | 准确率=19.20% | 耗时=3.87s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#8 | 损失=0.1032 | 准确率=19.32% | 耗时=2.09s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#0 | 损失=0.1224 | 准确率=19.24% | 耗时=2.97s | 上传=完整权重

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#1 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#5 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#4 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#8 | 加密耗时=0.04s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#0 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　Expansion 16.83x | plain=867.70 KB | cipher=14.26 MB

INFO:ckks_fed.log_utils:[聚合/AGG]　　Aggregating 5 encrypted updates...

INFO:ckks_fed.log_utils:[聚合/AGG]　　参与=5人 | 保留=5人 | 规则=CKKS+fedavg | 聚合耗时=0.034s | 上传总量=14.26 MB | 下载总量=867.70 KB

INFO:ckks_fed.log_utils:[服务器/SERVER] 全局模型已更新 | 当前轮次=4 | 上传模式=完整权重替换

INFO:ckks_fed.log_utils:[服务器/SERVER] 密文增量已准备 | 大小=2.85 MB | 加密耗时=0.0777s

INFO:ckks_fed.log_utils:[评估/EVAL]　　测试损失=0.1855 | 测试准确率=94.18%

INFO:ckks_fed.log_utils:[轮次/ROUND]　#4 done | acc=94.18% | round_dur=67.47s | vclock=4.50 min | comm=15.11 MB

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════ 第 5/20 轮 | CipherFetch 预取 ═══════════════════════════════════

INFO:ckks_fed.log_utils:[粘采样/STICKY] Select | target=4 | no pool, fallback random=[0, 5, 2, 6, 3]

INFO:ckks_fed.log_utils:[选择/SELECT] Presampled-Current 选中客户端: [0, 5, 2, 6, 3] (共5/10个)

INFO:ckks_fed.log_utils: 🔮 预取调度：为第 8 轮提前安排下载计划

INFO:ckks_fed.log_utils:[粘采样/STICKY] Presample | target=7 | sampled=5 (need=5, OC=1.0) | ids=[4, 8, 1, 3, 0]

INFO:ckks_fed.log_utils:[调度/SCHED] 当前轮=4 目标轮=7 | 最早启动轮=r_min(4) 可延后=0人 需立即启动=5人 | (OC=1.0, β=0.5)

INFO:ckks_fed.log_utils:[预取/PREFETCH] Broadcast → 5 clients | targets=[1, 4, 7, 8, 9]

INFO:ckks_fed.log_utils:[预取/PREFETCH] 广播完成 | 3 个空闲客户端已累加密文增量 | 平均缓存 1 个 delta (池内 3/5)

INFO:ckks_fed.log_utils: 💻 本地训练阶段（Fetch → Apply → Compute → Upload）

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#0 | 损失=0.0860 | 准确率=19.47% | 耗时=2.80s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#5 | 损失=0.0600 | 准确率=19.63% | 耗时=5.70s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #2 apply_prefetch | deltas=1 | apply_time=0.0152s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#2 | 损失=0.0887 | 准确率=19.45% | 耗时=2.72s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #6 apply_prefetch | deltas=1 | apply_time=0.0186s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#6 | 损失=0.0680 | 准确率=19.61% | 耗时=3.90s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #3 apply_prefetch | deltas=2 | apply_time=0.0173s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#3 | 损失=0.1124 | 准确率=19.48% | 耗时=1.51s | 上传=完整权重

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#0 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#5 | 加密耗时=0.05s | 密文大小=2.86 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#2 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#6 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#3 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] Expansion 16.83x | plain=867.70 KB | cipher=14.26 MB

INFO:ckks_fed.log_utils:[聚合/AGG] Aggregating 5 encrypted updates...

INFO:ckks_fed.log_utils:[聚合/AGG] 参与=5人 | 保留=5人 | 规则=CKKS+fedavg | 聚合耗时=0.035s | 上传总量=14.26 MB | 下载总量=867.70 KB

INFO:ckks_fed.log_utils:[服务器/SERVER] 全局模型已更新 | 当前轮次=5 | 上传模式=完整权重替换

INFO:ckks_fed.log_utils:[服务器/SERVER] 密文增量已准备 | 大小=2.85 MB | 加密耗时=0.0750s

INFO:ckks_fed.log_utils:[评估/EVAL] 测试损失=0.1564 | 测试准确率=94.96%

INFO:ckks_fed.log_utils:[轮次/ROUND] #5 done | acc=94.96% | round_dur=66.56s | vclock=5.61 min | comm=15.11 MB

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════ 第 6/20 轮 | CipherFetch 预取 ═══════════════════════════════════

INFO:ckks_fed.log_utils:[粘采样/STICKY] Select | target=5 | selected=[1, 6, 8, 7, 2] | pool=[1, 7, 6, 2, 8]

INFO:ckks_fed.log_utils:[选择/SELECT] Presampled-Current 选中客户端: [1, 6, 8, 7, 2] (共5/10个)

INFO:ckks_fed.log_utils: 🔮 预取调度：为第 9 轮提前安排下载计划

INFO:ckks_fed.log_utils:[粘采样/STICKY] Presample | target=8 | sampled=5 (need=5, OC=1.0) | ids=[0, 2, 6, 8, 7]

INFO:ckks_fed.log_utils:[调度/SCHED] 当前轮=5 目标轮=8 | 最早启动轮=r_min(5) 可延后=0人 需立即启动=5人 | (OC=1.0, β=0.5)

INFO:ckks_fed.log_utils:[预取/PREFETCH] Broadcast → 5 clients | targets=[0, 3, 4, 5, 9]

INFO:ckks_fed.log_utils:[预取/PREFETCH] 广播完成 | 1 个空闲客户端已累加密文增量 | 平均缓存 1 个 delta (池内 1/5)

INFO:ckks_fed.log_utils: 💻 本地训练阶段（Fetch → Apply → Compute → Upload）

INFO:ckks_fed.log_utils:[客户端/CLIENT] #1 apply_prefetch | deltas=1 | apply_time=0.0176s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#1 | 损失=0.0629 | 准确率=19.65% | 耗时=4.10s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#6 | 损失=0.0572 | 准确率=19.67% | 耗时=3.77s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #8 apply_prefetch | deltas=1 | apply_time=0.0167s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#8 | 损失=0.0606 | 准确率=19.64% | 耗时=2.52s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #7 apply_prefetch | deltas=4 | apply_time=0.0165s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#7 | 损失=0.1543 | 准确率=19.45% | 耗时=3.63s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#2 | 损失=0.0720 | 准确率=19.52% | 耗时=2.68s | 上传=完整权重

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#1 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#6 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#8 | 加密耗时=0.04s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#7 | 加密耗时=0.05s | 密文大小=2.86 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#2 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] Expansion 16.83x | plain=867.70 KB | cipher=14.26 MB

INFO:ckks_fed.log_utils:[聚合/AGG] Aggregating 5 encrypted updates...

INFO:ckks_fed.log_utils:[聚合/AGG] 参与=5人 | 保留=5人 | 规则=CKKS+fedavg | 聚合耗时=0.034s | 上传总量=14.26 MB | 下载总量=867.70 KB

INFO:ckks_fed.log_utils:[服务器/SERVER] 全局模型已更新 | 当前轮次=6 | 上传模式=完整权重替换

INFO:ckks_fed.log_utils:[服务器/SERVER] 密文增量已准备 | 大小=2.85 MB | 加密耗时=0.0774s

INFO:ckks_fed.log_utils:[评估/EVAL] 测试损失=0.1628 | 测试准确率=95.42%

INFO:ckks_fed.log_utils:[轮次/ROUND] #6 done | acc=95.42% | round_dur=59.48s | vclock=6.60 min | comm=15.11 MB

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════ 第 7/20 轮 | CipherFetch 预取 ═══════════════════════════════════

INFO:ckks_fed.log_utils:[粘采样/STICKY] Select | target=6 | selected=[8, 3, 5, 2, 7] | pool=[8, 2, 5, 7, 3]

INFO:ckks_fed.log_utils:[选择/SELECT] Presampled-Current 选中客户端: [8, 3, 5, 2, 7] (共5/10个)

INFO:ckks_fed.log_utils: 🔮 预取调度：为第 10 轮提前安排下载计划

INFO:ckks_fed.log_utils:[粘采样/STICKY] Presample | target=9 | sampled=5 (need=5, OC=1.0) | ids=[2, 1, 8, 9, 4]

INFO:ckks_fed.log_utils:[调度/SCHED] 当前轮=6 目标轮=9 | 最早启动轮=r_min(6) 可延后=0人 需立即启动=5人 | (OC=1.0, β=0.5)

INFO:ckks_fed.log_utils:[预取/PREFETCH] Broadcast → 5 clients | targets=[0, 1, 4, 6, 9]

INFO:ckks_fed.log_utils:[预取/PREFETCH] 广播完成 | 3 个空闲客户端已累加密文增量 | 平均缓存 1 个 delta (池内 3/5)

INFO:ckks_fed.log_utils: 💻 本地训练阶段（Fetch → Apply → Compute → Upload）

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#8 | 损失=0.0484 | 准确率=19.71% | 耗时=2.71s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #3 apply_prefetch | deltas=1 | apply_time=0.0178s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#3 | 损失=0.0744 | 准确率=19.64% | 耗时=1.80s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #5 apply_prefetch | deltas=1 | apply_time=0.0172s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#5 | 损失=0.0464 | 准确率=19.73% | 耗时=5.72s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#2 | 损失=0.0552 | 准确率=19.67% | 耗时=2.60s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#7 | 损失=0.0625 | 准确率=19.60% | 耗时=3.68s | 上传=完整权重

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#8 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#3 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#5 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#2 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#7 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] Expansion 16.82x | plain=867.70 KB | cipher=14.26 MB

INFO:ckks_fed.log_utils:[聚合/AGG] Aggregating 5 encrypted updates...

INFO:ckks_fed.log_utils:[聚合/AGG] 参与=5人 | 保留=5人 | 规则=CKKS+fedavg | 聚合耗时=0.035s | 上传总量=14.26 MB | 下载总量=867.70 KB

INFO:ckks_fed.log_utils:[服务器/SERVER] 全局模型已更新 | 当前轮次=7 | 上传模式=完整权重替换

INFO:ckks_fed.log_utils:[服务器/SERVER] 密文增量已准备 | 大小=2.85 MB | 加密耗时=0.0758s

INFO:ckks_fed.log_utils:[评估/EVAL] 测试损失=0.1436 | 测试准确率=95.91%

INFO:ckks_fed.log_utils:[轮次/ROUND] #7 done | acc=95.91% | round_dur=62.41s | vclock=7.64 min | comm=15.10 MB

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════ 第 8/20 轮 | CipherFetch 预取 ═══════════════════════════════════

INFO:ckks_fed.log_utils:[粘采样/STICKY] Select | target=7 | selected=[1, 4, 8, 3, 0] | pool=[4, 8, 1, 3, 0]

INFO:ckks_fed.log_utils:[选择/SELECT] Presampled-Current 选中客户端: [1, 4, 8, 3, 0] (共5/10个)

INFO:ckks_fed.log_utils: 🔮 预取调度：为第 11 轮提前安排下载计划

INFO:ckks_fed.log_utils:[粘采样/STICKY] Presample | target=10 | sampled=5 (need=5, OC=1.0) | ids=[1, 4, 2, 3, 5]

INFO:ckks_fed.log_utils:[调度/SCHED] 当前轮=7 目标轮=10 | 最早启动轮=r_min(7) 可延后=0人 需立即启动=5人 | (OC=1.0, β=0.5)

INFO:ckks_fed.log_utils:[预取/PREFETCH] Broadcast → 5 clients | targets=[2, 5, 6, 7, 9]

INFO:ckks_fed.log_utils:[预取/PREFETCH] 广播完成 | 2 个空闲客户端已累加密文增量 | 平均缓存 1 个 delta (池内 2/5)

INFO:ckks_fed.log_utils: 💻 本地训练阶段（Fetch → Apply → Compute → Upload）

INFO:ckks_fed.log_utils:[客户端/CLIENT] #1 apply_prefetch | deltas=1 | apply_time=0.0162s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#1 | 损失=0.0513 | 准确率=19.71% | 耗时=4.25s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #4 apply_prefetch | deltas=3 | apply_time=0.0161s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#4 | 损失=0.1042 | 准确率=19.56% | 耗时=4.33s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#8 | 损失=0.0374 | 准确率=19.77% | 耗时=2.20s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#3 | 损失=0.0571 | 准确率=19.69% | 耗时=1.80s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #0 apply_prefetch | deltas=2 | apply_time=0.0192s

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#0 | 损失=0.0573 | 准确率=19.72% | 耗时=3.09s | 上传=完整权重

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#1 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#4 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#8 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#3 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#0 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　Expansion 16.82x | plain=867.70 KB | cipher=14.25 MB

INFO:ckks_fed.log_utils:[聚合/AGG]　Aggregating 5 encrypted updates...

INFO:ckks_fed.log_utils:[聚合/AGG]　参与=5人 | 保留=5人 | 规则=CKKS+fedavg | 聚合耗时=0.035s | 上传总量=14.25 MB | 下载总量=867.70 KB

INFO:ckks_fed.log_utils:[服务器/SERVER] 全局模型已更新 | 当前轮次=8 | 上传模式=完整权重替换

INFO:ckks_fed.log_utils:[服务器/SERVER] 密文增量已准备 | 大小=2.85 MB | 加密耗时=0.0790s

INFO:ckks_fed.log_utils:[评估/EVAL]　测试损失=0.1444 | 测试准确率=96.47%

INFO:ckks_fed.log_utils:[轮次/ROUND]　#8 done | acc=96.47% | round_dur=67.90s | vclock=8.77 min | comm=15.10 MB

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════ 第 9/20 轮 | CipherFetch 预取 ═══════════════════════════════

INFO:ckks_fed.log_utils:[粘采样/STICKY] Select | target=8 | selected=[2, 8, 7, 0, 6] | pool=[0, 2, 6, 8, 7]

INFO:ckks_fed.log_utils:[选择/SELECT]　Presampled-Current 选中客户端: [2, 8, 7, 0, 6] (共5/10个)

INFO:ckks_fed.log_utils: 🔮 预取调度：为第 12 轮提前安排下载计划

INFO:ckks_fed.log_utils:[粘采样/STICKY] Presample | target=11 | sampled=5 (need=5, OC=1.0) | ids=[5, 8, 2, 3, 9]

INFO:ckks_fed.log_utils:[调度/SCHED]　当前轮=8 目标轮=11 | 最早启动轮=r_min(8) 可延后=1人 需立即启动=4人 | (OC=1.0, β=0.5)

INFO:ckks_fed.log_utils:[预取/PREFETCH] Broadcast → 5 clients | targets=[1, 3, 4, 5, 9]

INFO:ckks_fed.log_utils:[预取/PREFETCH] 广播完成 | 3 个空闲客户端已累加密文增量 | 平均缓存 1 个 delta (池内 3/5)

INFO:ckks_fed.log_utils: 💻 本地训练阶段（Fetch → Apply → Compute → Upload）

INFO:ckks_fed.log_utils:[客户端/CLIENT] #2 apply_prefetch | deltas=1 | apply_time=0.0167s

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#2 | 损失=0.0416 | 准确率=19.77% | 耗时=2.81s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#8 | 损失=0.0306 | 准确率=19.81% | 耗时=2.25s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #7 apply_prefetch | deltas=1 | apply_time=0.0176s

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#7 | 损失=0.0549 | 准确率=19.75% | 耗时=3.74s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#0 | 损失=0.0425 | 准确率=19.77% | 耗时=2.85s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #6 apply_prefetch | deltas=2 | apply_time=0.0166s

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#6 | 损失=0.0650 | 准确率=19.74% | 耗时=3.61s | 上传=完整权重

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#2 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#8 | 加密耗时=0.04s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#7 | 加密耗时=0.04s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#0 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#6 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　Expansion 16.82x | plain=867.70 KB | cipher=14.25 MB

INFO:ckks_fed.log_utils:[聚合/AGG]　Aggregating 5 encrypted updates...

INFO:ckks_fed.log_utils:[聚合/AGG]　参与=5人 | 保留=5人 | 规则=CKKS+fedavg | 聚合耗时=0.033s | 上传总量=14.25 MB | 下载总量=867.70 KB

INFO:ckks_fed.log_utils:[服务器/SERVER] 全局模型已更新 | 当前轮次=9 | 上传模式=完整权重替换

INFO:ckks_fed.log_utils:[服务器/SERVER] 密文增量已准备 | 大小=2.85 MB | 加密耗时=0.0751s

INFO:ckks_fed.log_utils:[评估/EVAL] 测试损失=0.1536 | 测试准确率=96.71%

INFO:ckks_fed.log_utils:[轮次/ROUND] #9 done | acc=96.71% | round_dur=66.67s | vclock=9.89 min | comm=15.10 MB

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════ 第 10/20 轮 | CipherFetch预取 ═══════════════════════════════════

INFO:ckks_fed.log_utils:[粘采样/STICKY] Select | target=9 | selected=[8, 4, 2, 1, 9] | pool=[2, 1, 8, 9, 4]

INFO:ckks_fed.log_utils:[选择/SELECT] Presampled-Current 选中客户端: [8, 4, 2, 1, 9] (共5/10个)

INFO:ckks_fed.log_utils: 🔮 预取调度：为第 13 轮提前安排下载计划

INFO:ckks_fed.log_utils:[粘采样/STICKY] Presample | target=12 | sampled=5 (need=5, OC=1.0) | ids=[7, 0, 1, 8, 3]

INFO:ckks_fed.log_utils:[调度/SCHED] 当前轮=9 目标轮=12 | 最早启动轮=r_min(9) 可延后=1人 需立即启动=4人 | (OC=1.0, β=0.5)

INFO:ckks_fed.log_utils:[预取/PREFETCH] Broadcast → 5 clients | targets=[0, 3, 5, 6, 7]

INFO:ckks_fed.log_utils:[预取/PREFETCH] 广播完成 | 3 个空闲客户端已累加密文增量 | 平均缓存 1 个 delta (池内 3/5)

INFO:ckks_fed.log_utils: 💻 本地训练阶段（Fetch → Apply → Compute → Upload）

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#8 | 损失=0.0242 | 准确率=19.87% | 耗时=2.12s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #4 apply_prefetch | deltas=1 | apply_time=0.0164s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#4 | 损失=0.0658 | 准确率=19.71% | 耗时=3.98s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#2 | 损失=0.0409 | 准确率=19.79% | 耗时=2.66s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #1 apply_prefetch | deltas=1 | apply_time=0.0172s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#1 | 损失=0.0531 | 准确率=19.76% | 耗时=4.35s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #9 apply_prefetch | deltas=6 | apply_time=0.0161s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#9 | 损失=0.8527 | 准确率=19.20% | 耗时=3.25s | 上传=完整权重

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#8 | 加密耗时=0.05s | 密文大小=2.86 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#4 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#2 | 加密耗时=0.04s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#1 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#9 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] Expansion 16.83x | plain=867.70 KB | cipher=14.26 MB

INFO:ckks_fed.log_utils:[聚合/AGG] Aggregating 5 encrypted updates...

INFO:ckks_fed.log_utils:[聚合/AGG] 参与=5人 | 保留=5人 | 规则=CKKS+fedavg | 聚合耗时=0.033s | 上传总量=14.26 MB | 下载总量=867.70 KB

INFO:ckks_fed.log_utils:[服务器/SERVER] 全局模型已更新 | 当前轮次=10 | 上传模式=完整权重替换

INFO:ckks_fed.log_utils:[服务器/SERVER] 密文增量已准备 | 大小=2.85 MB | 加密耗时=0.0726s

INFO:ckks_fed.log_utils:[评估/EVAL] 测试损失=0.4459 | 测试准确率=93.46%

INFO:ckks_fed.log_utils:[轮次/ROUND] #10 done | acc=93.46% | round_dur=68.07s | vclock=11.02 min | comm=15.11 MB

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════ 第 11/20 轮 | CipherFetch预取 ═══════════════════════════════

INFO:ckks_fed.log_utils:[粘采样/STICKY] Select | target=10 | selected=[4, 3, 5, 1, 2] | pool=[1, 4, 2, 3, 5]

INFO:ckks_fed.log_utils:[选择/SELECT] Presampled-Current 选中客户端: [4, 3, 5, 1, 2] (共5/10个)

INFO:ckks_fed.log_utils: 🔮 预取调度：为第 14 轮提前安排下载计划

INFO:ckks_fed.log_utils:[粘采样/STICKY] Presample | target=13 | sampled=5 (need=5, OC=1.0) | ids=[6, 3, 7, 8, 9]

INFO:ckks_fed.log_utils:[调度/SCHED] 当前轮=10 目标轮=13 | 最早启动轮=r_min(10) 可延后=2人 需立即启动=3人 | (OC=1.0, β=0.5)

INFO:ckks_fed.log_utils:[预取/PREFETCH] Broadcast → 5 clients | targets=[0, 6, 7, 8, 9]

INFO:ckks_fed.log_utils:[预取/PREFETCH] 广播完成 | 4 个空闲客户端已累加密文增量 | 平均缓存 1 个 delta (池内 4/5)

INFO:ckks_fed.log_utils: 💻 本地训练阶段（Fetch → Apply → Compute → Upload）

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#4 | 损失=0.0629 | 准确率=19.70% | 耗时=3.80s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #3 apply_prefetch | deltas=2 | apply_time=0.0158s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#3 | 损失=0.1356 | 准确率=19.68% | 耗时=1.54s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #5 apply_prefetch | deltas=3 | apply_time=0.0178s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#5 | 损失=0.1913 | 准确率=19.66% | 耗时=6.08s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#1 | 损失=0.0485 | 准确率=19.77% | 耗时=4.39s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#2 | 损失=0.0580 | 准确率=19.75% | 耗时=2.59s | 上传=完整权重

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#4 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#3 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#5 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#1 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#2 | 加密耗时=0.05s | 密文大小=2.86 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] Expansion 16.83x | plain=867.70 KB | cipher=14.26 MB

INFO:ckks_fed.log_utils:[聚合/AGG] Aggregating 5 encrypted updates...

INFO:ckks_fed.log_utils:[聚合/AGG] 参与=5人 | 保留=5人 | 规则=CKKS+fedavg | 聚合耗时=0.036s | 上传总量=14.26 MB | 下载总量=867.70 KB

INFO:ckks_fed.log_utils:[服务器/SERVER] 全局模型已更新 | 当前轮次=11 | 上传模式=完整权重替换

INFO:ckks_fed.log_utils:[服务器/SERVER] 密文增量已准备 | 大小=2.86 MB | 加密耗时=0.0820s

INFO:ckks_fed.log_utils:[评估/EVAL] 测试损失=0.4611 | 测试准确率=94.41%

INFO:ckks_fed.log_utils:[轮次/ROUND] #11 done | acc=94.41% | round_dur=67.36s | vclock=12.14 min | comm=15.11 MB

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════ 第 12/20 轮 | CipherFetch预取 ═══════════════════════════════════

INFO:ckks_fed.log_utils:[粘采样/STICKY] Select | target=11 | selected=[3, 2, 5, 9, 8] | pool=[5, 8, 2, 3, 9]

INFO:ckks_fed.log_utils:[选择/SELECT] Presampled-Current 选中客户端: [3, 2, 5, 9, 8] (共5/10个)

INFO:ckks_fed.log_utils: 🔮 预取调度：为第 15 轮提前安排下载计划

INFO:ckks_fed.log_utils:[粘采样/STICKY] Presample | target=14 | sampled=5 (need=5, OC=1.0) | ids=[1, 8, 5, 2, 9]

INFO:ckks_fed.log_utils:[调度/SCHED] 当前轮=11 目标轮=14 | 最早启动轮=r_min(11) 可延后=2人 需立即启动=3人 | (OC=1.0, β=0.5)

INFO:ckks_fed.log_utils:[预取/PREFETCH] Broadcast → 5 clients | targets=[0, 1, 4, 6, 7]

INFO:ckks_fed.log_utils:[预取/PREFETCH] 广播完成 | 1 个空闲客户端已累加密文增量 | 平均缓存 1 个 delta (池内 1/5)

INFO:ckks_fed.log_utils: 💻 本地训练阶段（Fetch → Apply → Compute → Upload）

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#3 | 损失=0.0486 | 准确率=19.83% | 耗时=1.42s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#2 | 损失=0.0586 | 准确率=19.79% | 耗时=2.17s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#5 | 损失=0.0350 | 准确率=19.86% | 耗时=5.81s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #9 apply_prefetch | deltas=1 | apply_time=0.0162s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#9 | 损失=0.0937 | 准确率=19.70% | 耗时=3.24s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #8 apply_prefetch | deltas=1 | apply_time=0.0167s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#8 | 损失=0.0821 | 准确率=19.75% | 耗时=3.36s | 上传=完整权重

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#3 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#2 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#5 | 加密耗时=0.05s | 密文大小=2.86 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#9 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#8 | 加密耗时=0.05s | 密文大小=2.86 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] Expansion 16.83x | plain=867.70 KB | cipher=14.26 MB

INFO:ckks_fed.log_utils:[聚合/AGG] Aggregating 5 encrypted updates...

INFO:ckks_fed.log_utils:[聚合/AGG] 参与=5人 | 保留=5人 | 规则=CKKS+fedavg | 聚合耗时=0.041s | 上传总量=14.26 MB | 下载总量=867.70 KB

INFO:ckks_fed.log_utils:[服务器/SERVER] 全局模型已更新 | 当前轮次=12 | 上传模式=完整权重替换

INFO:ckks_fed.log_utils:[服务器/SERVER] 密文增量已准备 | 大小=2.85 MB | 加密耗时=0.0745s

INFO:ckks_fed.log_utils:[评估/EVAL] 测试损失=0.4251 | 测试准确率=95.02%

INFO:ckks_fed.log_utils:[轮次/ROUND] #12 done | acc=95.02% | round_dur=68.00s | vclock=13.28 min | comm=15.11 MB

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════ 第 13/20 轮 | CipherFetch预取 ═══════════════════════════════════

INFO:ckks_fed.log_utils:[粘采样/STICKY] Select | target=12 | selected=[3, 0, 8, 7, 1] | pool=[7, 0, 1, 8, 3]

INFO:ckks_fed.log_utils:[选择/SELECT] Presampled-Current 选中客户端: [3, 0, 8, 7, 1] (共5/10个)

INFO:ckks_fed.log_utils: 🔮 预取调度：为第 16 轮提前安排下载计划

INFO:ckks_fed.log_utils:[粘采样/STICKY] Presample | target=15 | sampled=5 (need=5, OC=1.0) | ids=[8, 9, 4, 0, 7]

INFO:ckks_fed.log_utils:[调度/SCHED] 当前轮=12 目标轮=15 | 最早启动轮=r_min(12) 可延后=2人 需立即启动=3人 | (OC=1.0, β=0.5)

INFO:ckks_fed.log_utils:[预取/PREFETCH] Broadcast → 5 clients | targets=[2, 4, 5, 6, 9]

INFO:ckks_fed.log_utils:[预取/PREFETCH] 广播完成 | 2 个空闲客户端已累加密文增量 | 平均缓存 1 个 delta (池内 2/5)

INFO:ckks_fed.log_utils: 💻 本地训练阶段（Fetch → Apply → Compute → Upload）

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#3 | 损失=0.0523 | 准确率=19.84% | 耗时=1.84s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #0 apply_prefetch | deltas=3 | apply_time=0.0170s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#0 | 损失=0.2215 | 准确率=19.59% | 耗时=3.10s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#8 | 损失=0.0998 | 准确率=19.77% | 耗时=2.30s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #7 apply_prefetch | deltas=3 | apply_time=0.0177s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#7 | 损失=0.2920 | 准确率=19.50% | 耗时=4.03s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #1 apply_prefetch | deltas=1 | apply_time=0.0176s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#1 | 损失=0.0872 | 准确率=19.76% | 耗时=3.94s | 上传=完整权重

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#3 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#0 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#8 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#7 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#1 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] Expansion 16.83x | plain=867.70 KB | cipher=14.26 MB

INFO:ckks_fed.log_utils:[聚合/AGG] Aggregating 5 encrypted updates...

INFO:ckks_fed.log_utils:[聚合/AGG] 参与=5人 | 保留=5人 | 规则=CKKS+fedavg | 聚合耗时=0.036s | 上传总量=14.26 MB | 下载总量=867.70 KB

INFO:ckks_fed.log_utils:[服务器/SERVER] 全局模型已更新 | 当前轮次=13 | 上传模式=完整权重替换

INFO:ckks_fed.log_utils:[服务器/SERVER] 密文增量已准备 | 大小=2.85 MB | 加密耗时=0.0889s

INFO:ckks_fed.log_utils:[评估/EVAL] 测试损失=0.4854 | 测试准确率=95.30%

INFO:ckks_fed.log_utils:[轮次/ROUND] #13 done | acc=95.30% | round_dur=66.94s | vclock=14.39 min | comm=15.11 MB

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════ 第 14/20 轮 | CipherFetch预取 ═══════════════════════════════

INFO:ckks_fed.log_utils:[粘采样/STICKY] Select | target=13 | selected=[6, 7, 8, 3, 9] | pool=[6, 3, 7, 8, 9]

INFO:ckks_fed.log_utils:[选择/SELECT] Presampled-Current 选中客户端: [6, 7, 8, 3, 9] (共5/10个)

INFO:ckks_fed.log_utils: 🔮 预取调度：为第 17 轮提前安排下载计划

INFO:ckks_fed.log_utils:[粘采样/STICKY] Presample | target=16 | sampled=5 (need=5, OC=1.0) | ids=[1, 2, 7, 4, 8]

INFO:ckks_fed.log_utils:[调度/SCHED] 当前轮=13 目标轮=16 | 最早启动轮=r_min(13) 可延后=1人 需立即启动=4人 | (OC=1.0, β=0.5)

INFO:ckks_fed.log_utils:[预取/PREFETCH] Broadcast → 5 clients | targets=[0, 1, 2, 4, 5]

INFO:ckks_fed.log_utils:[预取/PREFETCH] 广播完成 | 3 个空闲客户端已累加密文增量 | 平均缓存 1 个 delta (池内 3/5)

INFO:ckks_fed.log_utils: 💻 本地训练阶段（Fetch → Apply → Compute → Upload）

INFO:ckks_fed.log_utils:[客户端/CLIENT] #6 apply_prefetch | deltas=4 | apply_time=0.0162s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#6 | 损失=3.0411 | 准确率=16.60% | 耗时=3.59s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#7 | 损失=0.0725 | 准确率=19.73% | 耗时=3.70s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#8 | 损失=0.0442 | 准确率=19.85% | 耗时=2.34s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#3 | 损失=0.0745 | 准确率=19.80% | 耗时=1.56s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #9 apply_prefetch | deltas=1 | apply_time=0.0172s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#9 | 损失=0.0790 | 准确率=19.79% | 耗时=3.47s | 上传=完整权重

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#6 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#7 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#8 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#3 | 加密耗时=0.05s | 密文大小=2.86 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#9 | 加密耗时=0.05s | 密文大小=2.86 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] Expansion 16.83x | plain=867.70 KB | cipher=14.26 MB

INFO:ckks_fed.log_utils:[聚合/AGG] Aggregating 5 encrypted updates...

INFO:ckks_fed.log_utils:[聚合/AGG] 参与=5人 | 保留=5人 | 规则=CKKS+fedavg | 聚合耗时=0.036s | 上传总量=14.26 MB | 下载总量=867.70 KB

INFO:ckks_fed.log_utils:[服务器/SERVER] 全局模型已更新 | 当前轮次=14 | 上传模式=完整权重替换

INFO:ckks_fed.log_utils:[服务器/SERVER] 密文增量已准备 | 大小=2.85 MB | 加密耗时=0.0757s

INFO:ckks_fed.log_utils:[评估/EVAL] 测试损失=2.3788 | 测试准确率=73.44%

INFO:ckks_fed.log_utils:[轮次/ROUND] #14 done | acc=73.44% | round_dur=68.40s | vclock=15.53 min | comm=15.11 MB

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════ 第 15/20 轮 | CipherFetch预取 ═══════════════════════════════════

INFO:ckks_fed.log_utils:[粘采样/STICKY] Select | target=14 | selected=[8, 5, 9, 2, 1] | pool=[1, 8, 5, 2, 9]

INFO:ckks_fed.log_utils:[选择/SELECT] Presampled-Current 选中客户端: [8, 5, 9, 2, 1] (共5/10个)

INFO:ckks_fed.log_utils: 🔮 预取调度：为第 18 轮提前安排下载计划

INFO:ckks_fed.log_utils:[粘采样/STICKY] Presample | target=17 | sampled=5 (need=5, OC=1.0) | ids=[3, 1, 8, 7, 4]

INFO:ckks_fed.log_utils:[调度/SCHED] 当前轮=14 目标轮=17 | 最早启动轮=r_min(14) 可延后=2人 需立即启动=3人 | (OC=1.0, β=0.5)

INFO:ckks_fed.log_utils:[预取/PREFETCH] Broadcast → 5 clients | targets=[0, 3, 4, 6, 7]

INFO:ckks_fed.log_utils:[预取/PREFETCH] 广播完成 | 3 个空闲客户端已累加密文增量 | 平均缓存 1 个 delta (池内 3/5)

INFO:ckks_fed.log_utils: 💻 本地训练阶段（Fetch → Apply → Compute → Upload）

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#8 | 损失=0.1795 | 准确率=19.37% | 耗时=2.52s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #5 apply_prefetch | deltas=2 | apply_time=0.0186s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#5 | 损失=0.5019 | 准确率=17.72% | 耗时=6.07s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#9 | 损失=0.1704 | 准确率=19.39% | 耗时=3.25s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #2 apply_prefetch | deltas=2 | apply_time=0.0166s

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#2 | 损失=0.5997 | 准确率=17.81% | 耗时=2.10s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #1 apply_prefetch | deltas=1 | apply_time=0.0157s

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#1 | 损失=0.2378 | 准确率=19.19% | 耗时=4.11s | 上传=完整权重

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#8 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#5 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#9 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#2 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#1 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　Expansion 16.82x | plain=867.70 KB | cipher=14.25 MB

INFO:ckks_fed.log_utils:[聚合/AGG]　　Aggregating 5 encrypted updates...

INFO:ckks_fed.log_utils:[聚合/AGG]　　参与=5人 | 保留=5人 | 规则=CKKS+fedavg | 聚合耗时=0.034s | 上传总量=14.25 MB | 下载总量=867.70 KB

INFO:ckks_fed.log_utils:[服务器/SERVER] 全局模型已更新 | 当前轮次=15 | 上传模式=完整权重替换

INFO:ckks_fed.log_utils:[服务器/SERVER] 密文增量已准备 | 大小=2.85 MB | 加密耗时=0.0816s

INFO:ckks_fed.log_utils:[评估/EVAL]　　测试损失=1.1215 | 测试准确率=79.62%

INFO:ckks_fed.log_utils:[轮次/ROUND]　#15 done | acc=79.62% | round_dur=68.10s | vclock=16.67 min | comm=15.10 MB

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════ 第 16/20 轮 | CipherFetch预取 ═══════════════════════════════════

INFO:ckks_fed.log_utils:[粘采样/STICKY] Select | target=15 | selected=[9, 8, 4, 7, 0] | pool=[8, 9, 4, 0, 7]

INFO:ckks_fed.log_utils:[选择/SELECT]　Presampled-Current 选中客户端: [9, 8, 4, 7, 0]（共5/10个）

INFO:ckks_fed.log_utils: 🔮 预取调度：为第 19 轮提前安排下载计划

INFO:ckks_fed.log_utils:[粘采样/STICKY] Presample | target=18 | sampled=5 (need=5, OC=1.0) | ids=[2, 4, 6, 1, 9]

INFO:ckks_fed.log_utils:[调度/SCHED]　当前轮=15 目标轮=18 | 最早启动轮=r_min(15) 可延后=3人 需立即启动=2人 | (OC=1.0, β=0.5)

INFO:ckks_fed.log_utils:[预取/PREFETCH] Broadcast → 5 clients | targets=[1, 2, 3, 5, 6]

INFO:ckks_fed.log_utils:[预取/PREFETCH] 广播完成 | 3 个空闲客户端已累加密文增量 | 平均缓存 1 个 delta（池内 3/5）

INFO:ckks_fed.log_utils: 💻 本地训练阶段（Fetch → Apply → Compute → Upload）

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#9 | 损失=0.2362 | 准确率=18.93% | 耗时=3.73s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#8 | 损失=0.2690 | 准确率=18.74% | 耗时=1.97s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #4 apply_prefetch | deltas=4 | apply_time=0.0161s

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#4 | 损失=0.6903 | 准确率=16.31% | 耗时=3.77s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #7 apply_prefetch | deltas=1 | apply_time=0.0167s

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#7 | 损失=0.3205 | 准确率=18.43% | 耗时=4.13s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #0 apply_prefetch | deltas=2 | apply_time=0.0189s

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#0 | 损失=0.5163 | 准确率=17.47% | 耗时=3.64s | 上传=完整权重

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#9 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#8 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#4 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#7 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#0 | 加密耗时=0.04s | 密文大小=2.86 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] Expansion 16.83x | plain=867.70 KB | cipher=14.26 MB

INFO:ckks_fed.log_utils:[聚合/AGG] Aggregating 5 encrypted updates...

INFO:ckks_fed.log_utils:[聚合/AGG] 参与=5人 | 保留=5人 | 规则=CKKS+fedavg | 聚合耗时=0.033s | 上传总量=14.26 MB | 下载总量=867.70 KB

INFO:ckks_fed.log_utils:[服务器/SERVER] 全局模型已更新 | 当前轮次=16 | 上传模式=完整权重替换

INFO:ckks_fed.log_utils:[服务器/SERVER] 密文增量已准备 | 大小=2.85 MB | 加密耗时=0.0766s

INFO:ckks_fed.log_utils:[评估/EVAL] 测试损失=0.9347 | 测试准确率=72.92%

INFO:ckks_fed.log_utils:[轮次/ROUND] #16 done | acc=72.92% | round_dur=68.50s | vclock=17.81 min | comm=15.11 MB

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════ 第 17/20 轮 | CipherFetch预取 ═══════════════════════════════════

INFO:ckks_fed.log_utils:[粘采样/STICKY] Select | target=16 | selected=[8, 4, 1, 7, 2] | pool=[1, 2, 7, 4, 8]

INFO:ckks_fed.log_utils:[选择/SELECT] Presampled-Current 选中客户端: [8, 4, 1, 7, 2] (共5/10个)

INFO:ckks_fed.log_utils: 🔮 预取调度：为第 20 轮提前安排下载计划

INFO:ckks_fed.log_utils:[粘采样/STICKY] Presample | target=19 | sampled=5 (need=5, OC=1.0) | ids=[6, 5, 3, 4, 1]

INFO:ckks_fed.log_utils:[调度/SCHED] 当前轮=16 目标轮=19 | 最早启动轮=r_min(16) 可延后=3人 需立即启动=2人 | (OC=1.0, β=0.5)

INFO:ckks_fed.log_utils:[预取/PREFETCH] Broadcast → 5 clients | targets=[0, 3, 5, 6, 9]

INFO:ckks_fed.log_utils:[预取/PREFETCH] 广播完成 | 3 个空闲客户端已累加密文增量 | 平均缓存 1 个 delta (池内 3/5)

INFO:ckks_fed.log_utils: 💻 本地训练阶段（Fetch → Apply → Compute → Upload）

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#8 | 损失=0.2497 | 准确率=18.62% | 耗时=2.55s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#4 | 损失=0.3151 | 准确率=18.21% | 耗时=4.03s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #1 apply_prefetch | deltas=1 | apply_time=0.0151s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#1 | 损失=0.3315 | 准确率=18.55% | 耗时=4.35s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#7 | 损失=0.2286 | 准确率=18.83% | 耗时=4.48s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #2 apply_prefetch | deltas=1 | apply_time=0.0225s

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#2 | 损失=0.4249 | 准确率=17.80% | 耗时=2.73s | 上传=完整权重

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#8 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#4 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#1 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#7 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#2 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　Expansion 16.83x | plain=867.70 KB | cipher=14.26 MB

INFO:ckks_fed.log_utils:[聚合/AGG]　Aggregating 5 encrypted updates...

INFO:ckks_fed.log_utils:[聚合/AGG]　参与=5人 | 保留=5人 | 规则=CKKS+fedavg | 聚合耗时=0.035s | 上传总量=14.26 MB | 下载总量=867.70 KB

INFO:ckks_fed.log_utils:[服务器/SERVER] 全局模型已更新 | 当前轮次=17 | 上传模式=完整权重替换

INFO:ckks_fed.log_utils:[服务器/SERVER] 密文增量已准备 | 大小=2.85 MB | 加密耗时=0.0850s

INFO:ckks_fed.log_utils:[评估/EVAL]　测试损失=0.7160 | 测试准确率=82.80%

INFO:ckks_fed.log_utils:[轮次/ROUND]　#17 done | acc=82.80% | round_dur=67.66s | vclock=18.94 min | comm=15.11 MB

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════ 第 18/20 轮 | CipherFetch预取 ═══════════════════════════════

INFO:ckks_fed.log_utils:[粘采样/STICKY] Select | target=17 | selected=[1, 8, 3, 7, 4] | pool=[3, 1, 8, 7, 4]

INFO:ckks_fed.log_utils:[选择/SELECT]　Presampled-Current 选中客户端: [1, 8, 3, 7, 4] (共5/10个)

INFO:ckks_fed.log_utils: 🔮 预取调度：为第 21 轮提前安排下载计划

INFO:ckks_fed.log_utils:[粘采样/STICKY] Presample | target=20 | sampled=5 (need=5, OC=1.0) | ids=[0, 5, 7, 9, 6]

INFO:ckks_fed.log_utils:[调度/SCHED] 当前轮=17 目标轮=20 | 最早启动轮=r_min(17) 可延后=2人 需立即启动=3人 | (OC=1.0, β=0.5)

INFO:ckks_fed.log_utils:[预取/PREFETCH] Broadcast → 5 clients | targets=[0, 2, 5, 6, 9]

INFO:ckks_fed.log_utils:[预取/PREFETCH] 广播完成 | 4 个空闲客户端已累加密文增量 | 平均缓存 1 个 delta (池内 4/5)

INFO:ckks_fed.log_utils: 💻 本地训练阶段（Fetch → Apply → Compute → Upload）

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#1 | 损失=0.2188 | 准确率=18.96% | 耗时=4.11s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#8 | 损失=0.2599 | 准确率=18.70% | 耗时=2.49s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #3 apply_prefetch | deltas=3 | apply_time=0.0172s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#3 | 损失=1.1005 | 准确率=14.09% | 耗时=1.70s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#7 | 损失=0.2013 | 准确率=19.00% | 耗时=4.07s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#4 | 损失=0.2716 | 准确率=18.41% | 耗时=4.25s | 上传=完整权重

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#1 | 加密耗时=0.04s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#8 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#3 | 加密耗时=0.04s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#7 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#4 | 加密耗时=0.05s | 密文大小=2.86 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] Expansion 16.83x | plain=867.70 KB | cipher=14.26 MB

INFO:ckks_fed.log_utils:[聚合/AGG] Aggregating 5 encrypted updates...

INFO:ckks_fed.log_utils:[聚合/AGG] 参与=5人 | 保留=5人 | 规则=CKKS+fedavg | 聚合耗时=0.034s | 上传总量=14.26 MB | 下载总量=867.70 KB

INFO:ckks_fed.log_utils:[服务器/SERVER] 全局模型已更新 | 当前轮次=18 | 上传模式=完整权重替换

INFO:ckks_fed.log_utils:[服务器/SERVER] 密文增量已准备 | 大小=2.85 MB | 加密耗时=0.0768s

INFO:ckks_fed.log_utils:[评估/EVAL] 测试损失=1.1388 | 测试准确率=81.06%

INFO:ckks_fed.log_utils:[轮次/ROUND] #18 done | acc=81.06% | round_dur=67.89s | vclock=20.07 min | comm=15.11 MB

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════ 第 19/20 轮 | CipherFetch预取 ═══════════════════════════════════

INFO:ckks_fed.log_utils:[粘采样/STICKY] Select | target=18 | selected=[1, 4, 9, 2, 6] | pool=[2, 4, 6, 1, 9]

INFO:ckks_fed.log_utils:[选择/SELECT] Presampled-Current 选中客户端: [1, 4, 9, 2, 6] (共5/10个)

INFO:ckks_fed.log_utils: 🔮 预取调度：为第 22 轮提前安排下载计划

INFO:ckks_fed.log_utils:[粘采样/STICKY] Presample | target=21 | sampled=5 (need=5, OC=1.0) | ids=[6, 9, 0, 8, 2]

INFO:ckks_fed.log_utils:[调度/SCHED] 当前轮=18 目标轮=21 | 最早启动轮=r_min(18) 可延后=3人 需立即启动=2人 | (OC=1.0, β=0.5)

INFO:ckks_fed.log_utils:[预取/PREFETCH] Broadcast → 5 clients | targets=[0, 3, 5, 7, 8]

INFO:ckks_fed.log_utils:[预取/PREFETCH] 广播完成 | 2 个空闲客户端已累加密文增量 | 平均缓存 1 个 delta (池内 2/5)

INFO:ckks_fed.log_utils: 💻 本地训练阶段（Fetch → Apply → Compute → Upload）

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#1 | 损失=0.1809 | 准确率=19.14% | 耗时=4.13s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#4 | 损失=0.2512 | 准确率=18.52% | 耗时=4.14s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #9 apply_prefetch | deltas=2 | apply_time=0.0162s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#9 | 损失=0.4188 | 准确率=18.35% | 耗时=3.55s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #2 apply_prefetch | deltas=1 | apply_time=0.0172s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#2 | 损失=0.3821 | 准确率=18.08% | 耗时=3.01s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #6 apply_prefetch | deltas=4 | apply_time=0.0152s

INFO:ckks_fed.log_utils:[训练/TRAIN]　客户端#6 | 损失=0.8250 | 准确率=15.94% | 耗时=4.01s | 上传=完整权重

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#1 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#4 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#9 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#2 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　客户端#6 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT]　Expansion 16.83x | plain=867.70 KB | cipher=14.26 MB

INFO:ckks_fed.log_utils:[聚合/AGG]　Aggregating 5 encrypted updates...

INFO:ckks_fed.log_utils:[聚合/AGG]　参与=5人 | 保留=5人 | 规则=CKKS+fedavg | 聚合耗时=0.034s | 上传总量=14.26 MB | 下载总量=867.70 KB

INFO:ckks_fed.log_utils:[服务器/SERVER]　全局模型已更新 | 当前轮次=19 | 上传模式=完整权重替换

INFO:ckks_fed.log_utils:[服务器/SERVER]　密文增量已准备 | 大小=2.85 MB | 加密耗时=0.0783s

INFO:ckks_fed.log_utils:[评估/EVAL]　测试损失=0.8807 | 测试准确率=75.11%

INFO:ckks_fed.log_utils:[轮次/ROUND]　#19 done | acc=75.11% | round_dur=68.40s | vclock=21.21 min | comm=15.11 MB

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════ 第 20/20 轮 | CipherFetch预取 ═══════════════════════════════════

INFO:ckks_fed.log_utils:[粘采样/STICKY]　Select | target=19 | selected=[6, 1, 5, 3, 4] | pool=[6, 5, 3, 4, 1]

INFO:ckks_fed.log_utils:[选择/SELECT]　Presampled-Current 选中客户端: [6, 1, 5, 3, 4] (共5/10个)

INFO:ckks_fed.log_utils: 🔮 预取调度：为第 23 轮提前安排下载计划

INFO:ckks_fed.log_utils:[粘采样/STICKY]　Presample | target=22 | sampled=5 (need=5, OC=1.0) | ids=[5, 0, 4, 1, 6]

INFO:ckks_fed.log_utils:[调度/SCHED]　当前轮=19 目标轮=22 | 最早启动轮=r_min(19) 可延后=2人 需立即启动=3人 | (OC=1.0, β=0.5)

INFO:ckks_fed.log_utils:[预取/PREFETCH] Broadcast → 5 clients | targets=[0, 2, 7, 8, 9]

INFO:ckks_fed.log_utils:[预取/PREFETCH] 广播完成 | 1 个空闲客户端已累加密文增量 | 平均缓存 3 个 delta (池内 1/5)

INFO:ckks_fed.log_utils: 💻 本地训练阶段（Fetch → Apply → Compute → Upload）

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#6 | 损失=0.3339 | 准确率=18.40% | 耗时=3.92s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#1 | 损失=0.2687 | 准确率=19.08% | 耗时=4.10s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #5 apply_prefetch | deltas=4 | apply_time=0.0162s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#5 | 损失=2.1141 | 准确率=6.69% | 耗时=6.35s | 上传=完整权重

INFO:ckks_fed.log_utils:[客户端/CLIENT] #3 apply_prefetch | deltas=1 | apply_time=0.0191s

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#3 | 损失=0.6055 | 准确率=17.11% | 耗时=1.98s | 上传=完整权重

INFO:ckks_fed.log_utils:[训练/TRAIN] 客户端#4 | 损失=0.4551 | 准确率=17.14% | 耗时=4.43s | 上传=完整权重

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#6 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#1 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#5 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#3 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] 客户端#4 | 加密耗时=0.05s | 密文大小=2.85 MB | 密文块数=19

INFO:ckks_fed.log_utils:[加密/ENCRYPT] Expansion 16.82x | plain=867.70 KB | cipher=14.26 MB

INFO:ckks_fed.log_utils:[聚合/AGG] Aggregating 5 encrypted updates...

INFO:ckks_fed.log_utils:[聚合/AGG] 参与=5人 | 保留=5人 | 规则=CKKS+fedavg | 聚合耗时=0.034s | 上传总量=14.26 MB | 下载总量=867.70 KB

INFO:ckks_fed.log_utils:[服务器/SERVER] 全局模型已更新 | 当前轮次=20 | 上传模式=完整权重替换

INFO:ckks_fed.log_utils:[服务器/SERVER] 密文增量已准备 | 大小=2.85 MB | 加密耗时=0.0776s

INFO:ckks_fed.log_utils:[评估/EVAL]　测试损失=1.2207 | 测试准确率=62.70%

INFO:ckks_fed.log_utils:[轮次/ROUND]　#20 done | acc=62.70% | round_dur=67.99s | vclock=22.34 min | comm=15.10 MB

INFO:**main**:指标已保存到: ./results\experiment_20260420_182609\metrics.json

INFO:**main**:Markdown报告已保存: ./results\experiment_20260420_182609\report.md

INFO:**main**:实验注册表已更新: ./results\experiments_registry.json

INFO:**main**:结果已保存到: ./results\experiment_20260420_182609

INFO:**main**:模型已保存: ./history_models\models_20260420_182609\model_round_20.pth

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════════ 客户端参与统计与带宽/下载量 ═══════════════════════════════════════

INFO:ckks_fed.log_utils:

Client	Part	CTs	DL BW	UL BW	Prefetch ↓	Fetch ↓	Total ↓
0	9	171	1.08 Mbps	366.1 kbps	14.25 MB	14.25 MB	28.50 MB
1	13	247	885.8 kbps	427.0 kbps	17.11 MB	2.85 MB	19.96 MB
2	11	209	1.14 Mbps	566.3 kbps	11.41 MB	8.55 MB	19.96 MB
3	10	190	1.24 Mbps	385.1 kbps	14.26 MB	11.40 MB	25.66 MB
4	9	171	976.0 kbps	367.6 kbps	14.25 MB	14.26 MB	28.51 MB
5	9	171	851.1 kbps	513.6 kbps	11.40 MB	19.95 MB	31.35 MB
6	7	133	733.1 kbps	419.5 kbps	11.40 MB	22.80 MB	34.20 MB

INFO:ckks_fed.log_utils:│ 7 │ 9 │ 171 │ 1.09 Mbps │ 525.8 kbps │ 14.25 MB │ 11.40 MB │ 25.65 MB │

INFO:ckks_fed.log_utils:│ 8 │ 16 │ 304 │ 852.2 kbps │ 539.4 kbps │ 8.55 MB │ 0 B │ 8.55 MB │

INFO:ckks_fed.log_utils:│ 9 │ 7 │ 133 │ 1.19 Mbps │ 360.4 kbps │ 14.25 MB │ 17.11 MB │ 31.35 MB │

INFO:ckks_fed.log_utils:└──────────┴──────────┴──────────┴──────────┴──────────┴──────────┴──────────┴──────────┘

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils: ⏱️ 虚拟时钟说明

INFO:ckks_fed.log_utils: ├── round_duration = 最慢选中客户端的 (补下载时间 + 训练时间 + 上传时间)

INFO:ckks_fed.log_utils: └── virtual_clock = 所有轮次 round_duration 的累加（模拟真实 wall-clock 训练时间）

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════════ 实验结束 ═══════════════════════════════════════

INFO:ckks_fed.log_utils:

INFO:ckks_fed.log_utils:═══════════════════════════════════════ 实验结果汇总 ═══════════════════════════════════════

INFO:ckks_fed.log_utils: 最终测试准确率: 62.70%

INFO:ckks_fed.log_utils: 总训练时间: 22.34 min

INFO:ckks_fed.log_utils: 总通信量: 302.11 MB

INFO:ckks_fed.log_utils: 密文膨胀率: 16.83x

INFO:ckks_fed.log_utils:──

INFO:ckks_fed.log_utils:[汇总/SUMMARY] 实验结果已保存至: ./results\experiment_20260420_182609

进程已结束，退出代码为 0